

# B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

## IT 19-204-0404 SOFTWARE ENGINEERING (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

### Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Apply engineering aspects required for building software in a systematic way.
- CO2: Formally specify the requirements and choose an appropriate life cycle model.
- CO3: Apply various design concepts and UML notations in their academic projects.
- CO4: Explain fundamental elements of Software Project Management.
- CO5: Estimate software project effort, cost and schedule for an intermediate size project.
- CO6: Identify and manage software project risks.
- CO7: Develop various test plans and apply in the academic projects.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

### PART A

(Answer ALL questions)

(8 × 3 = 24)

|                                                                                                                                    | Marks | BL | CO | PO   |
|------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|------|
| I. (a) List the major differences between the exploratory and modern software development practices.                               | 3     | L1 | 1  | 1,2  |
| (b) Identify an inconsistent requirement. Explain why you think that the requirement is inconsistent.                              | 3     | L3 | 2  | 2    |
| (c) What do you mean by modular design? How can you determine whether a given design is modular or not?                            | 3     | L4 | 3  | 1    |
| (d) Why we need to develop use case diagrams in software development?                                                              | 3     | L2 | 3  | 1,4  |
| (e) What is TQM (Total Quality Management)?                                                                                        | 3     | L1 | 4  | 3    |
| (f) What is system testing? What are the different kinds of system testing that are usually performed on a large software product? | 3     | L3 | 7  | 1,2  |
| (g) Explain the application of activity network in project management.                                                             | 3     | L2 | 5  | 3    |
| (h) Distinguish between functional and project organization structures.                                                            | 3     | L1 | 5  | 1,12 |

### PART B

(4 × 12 = 48)

- II. Briefly explain the agile software development model. Give an example of a project for which the agile model would be suitable and one project for which the agile model would not be appropriate.
- OR
- III. Explain the organization of SRS as prescribed by IEEE. Distinguish functional and non-functional requirements. Identify functional and non-functional requirements for a library management software.
- IV. Give a detailed explanation of the structured analysis and structured design methods. Explain the significance of the structure chart and DFD in SA/SD using the appropriate examples and illustrations.
- OR
- V. Explain the terms coupling and cohesion in relation to software design. Give an explanation of each classification along with examples.

(P.T.O.)

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|       |                                                                                                                                                                                                                                                                                                                                      | Marks | BL | CO | PO   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|------|
| VI.   | Explain White box testing approaches. How these White box testing approaches can be used for test case design?                                                                                                                                                                                                                       | 12    | L2 | 7  | 1,4  |
|       | OR                                                                                                                                                                                                                                                                                                                                   |       |    |    |      |
| VII.  | (a) Explain any four quality factors associated with a software product.                                                                                                                                                                                                                                                             | 4     | L2 | 4  | 1,12 |
|       | (b) Explain the five levels of SEI CMM by clearly specifying focus areas and key process areas of each level.                                                                                                                                                                                                                        | 8     | L2 | 4  | 1,12 |
|       |                                                                                                                                                                                                                                                                                                                                      |       |    |    |      |
| VIII. | (a) Explain function point metric for software size estimation.                                                                                                                                                                                                                                                                      | 4     | L2 | 5  | 1,2  |
|       | (b) Differentiate between basic COCOMO and intermediate COCOMO. Apply the basic COCOMO estimation model for an organic type of software project to estimate effort and time. Assume that the size of the software product has been estimated to be 32000 lines of source code ( $a = 2.4$ , $b = 1.05$ , $c = 2.5$ and $d = 0.38$ ). | 8     | L3 | 5  | 1,2  |
|       | OR                                                                                                                                                                                                                                                                                                                                   |       |    |    |      |
| IX.   | (a) List three common types of risks that a typical software project might suffer from.                                                                                                                                                                                                                                              | 3     | L2 | 6  | 2,12 |
|       | (b) Explain how you can identify the risks that your project is susceptible to. Suppose you are the project manager of a large software development project, point out the main steps you would follow to manage risks in your software project.                                                                                     | 9     | L3 | 6  | 2,12 |

Blooms's Taxonomy Level's

L1 – 12.5%, L2 – 47.23%, L3 – 36.11%, L4 – 4.16%.

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